

Classwork 6 Solutions /* Revised */

2025/04/09

1. Using T flip-flops, design a 3-bit counter which counts in the following sequence. Provide the state table, transition table, and next state equations for the T flip-flops.

001 → 011 → 010 → 110 → 111 → 101 → 100 → 000 → 001 → ...

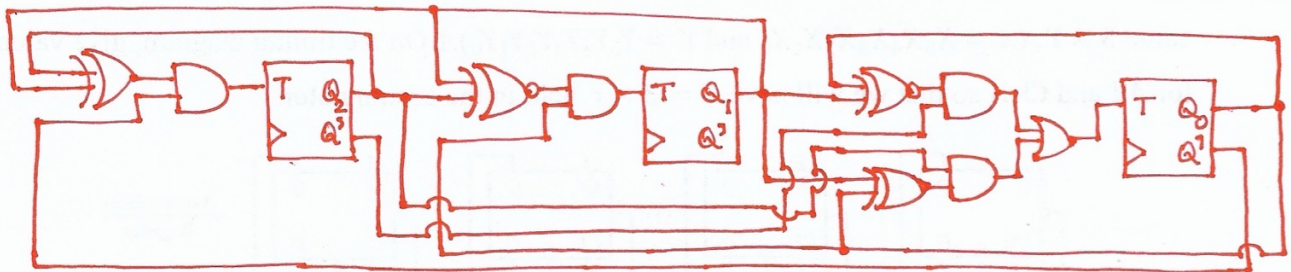
#	Present State	Next State	$Q_2 Q_1 Q_0$	$Q_2' Q_1' Q_0'$	$T_2 T_1 T_0$
1	001	011	0 0 1	0 0 1	0 0 1
2	011	010	0 0 1	0 1 1	0 1 0
3	010	110	0 1 0	1 1 0	1 0 0
4	110	111	0 1 1	0 1 0	0 0 1
5	111	101	1 0 0	0 0 0	1 0 0
6	101	100	1 0 1	1 0 0	0 0 1
7	100	000	1 1 0	1 1 1	0 0 1
8	000	001	1 1 1	1 0 1	0 1 0

$Q_2 \backslash Q_1 Q_0$	00	01	11	10
0	0	0	0	1
1	1	0	0	0

$Q_2 \backslash Q_1 Q_0$	00	01	11	10
0	0	1	0	0
1	0	0	1	0

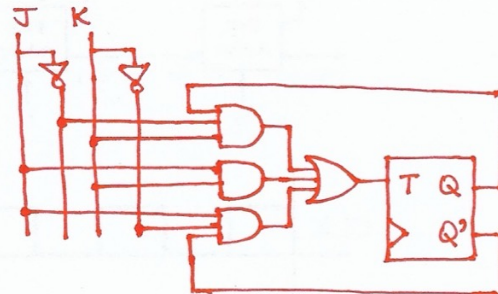
$Q_2 \backslash Q_1 Q_0$	00	01	11	10
0	1	0	1	0
1	0	1	0	1

$$\begin{aligned}
 T_2 &= Q_2' Q_1 Q_0' + Q_2 Q_1' Q_0' \\
 &= (Q_2' Q_1 + Q_2 Q_1') Q_0' \\
 &= (Q_2 \oplus Q_1) Q_0' \\
 T_1 &= Q_2' Q_1' Q_0 + Q_2 Q_1 Q_0 \\
 &= (Q_2' Q_1' + Q_2 Q_1) Q_0 \\
 &= (Q_2 \equiv Q_1) Q_0 \\
 T_0 &= Q_2' Q_1' Q_0' + Q_2' Q_1 Q_0 \\
 &\quad + Q_2 Q_1' Q_0 + Q_2 Q_1 Q_0' \\
 &= Q_2' (Q_1 \equiv Q_0) + Q_2 (Q_1 \oplus Q_0)
 \end{aligned}$$

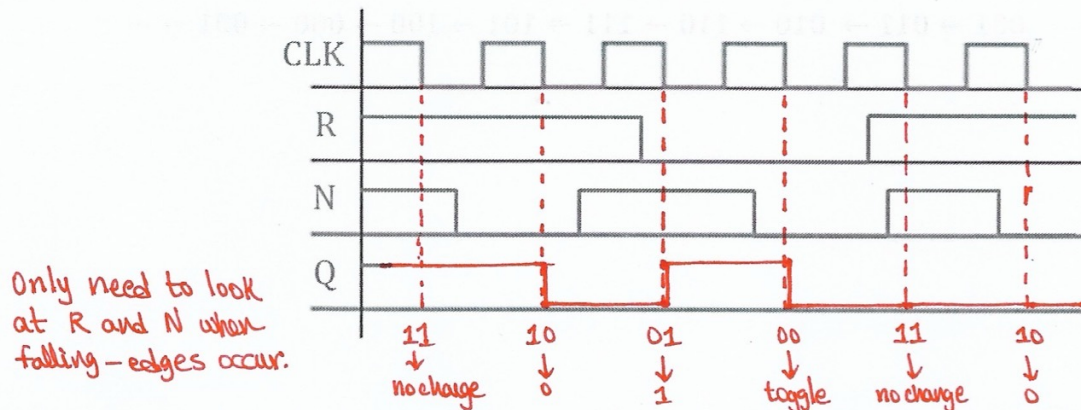


2. Show how a T flip-flop can be converted to a J-K flip-flop by adding external gates.

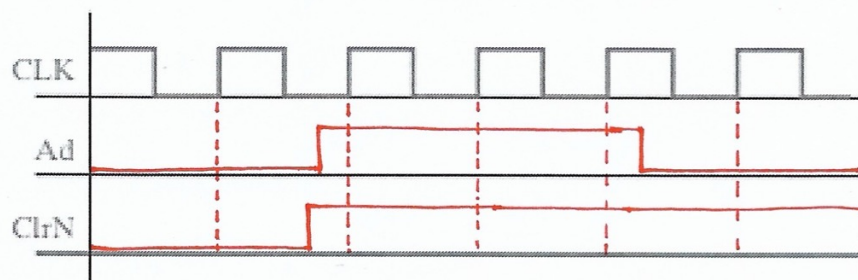
$$\begin{aligned}
 \text{JK: } \begin{cases} 00 : \text{no change} & \Rightarrow T = 0 \\ 01 : 0 & \Rightarrow T = Q \\ 10 : 1 & \Rightarrow T = Q' \\ 11 : \text{toggle} & \Rightarrow T = 1 \end{cases}
 \end{aligned}$$



- Complete the given timing diagram by assuming negligible propagation delay.



- [illegible]



ClrN is used to asynchronously reset/clear the unwanted value from X. Ad needs to be high for exactly 3 rising edges of the clock, sometime after clearing the previous value so that Y can be accumulated, three (3) times such that $X = 3 \times Y$ at the end.